Mathematical Statistics — Solutions

Graduate entrance practice problems

Contents

A. Analysis and linear algebra **1**

B. Distributions and transformations **2**

C. Convergence and limit theorems **7**

D. Point estimation **9**

E. Hypothesis testing **15**

F. Interval estimation and Bayesian inference **20**

G. Nonparametric methods **20**

A. Analysis and linear algebra

Solution 1 *Analysis*

(a) Apply the MVT to $\log(1+t)$ on $[0,x]$: $\log(1+x) = \frac{x}{1+\xi}$ for some $\xi \in (0,x)$. Since $1 < 1+\xi < 1+x$, we get $\frac{x}{1+x} < \log(1+x) < x$.

(b) $f_n(0) = 0$, and for $x > 0$, $nxe^{-nx} \to 0$ because $ue^{-u} \to 0$ as $u \to \infty$. So $f \equiv 0$. Since $\frac{d}{dx} f_n = ne^{-nx}(1-nx)$, f_n peaks at $x = 1/n$ with $\sup_{x\geq 0} f_n = f_n(1/n) = e^{-1} \not\to 0$: not uniform on $[0,\infty)$. On $[a,\infty)$, for $n > 1/a$ the function is decreasing there, so $\sup_{x\geq a} f_n = nae^{-na} \to 0$: uniform.

(c) $\int_0^1 nxe^{-nx}dx = \frac{1}{n}\int_0^n ue^{-u}du \leq \frac{1}{n}\int_0^\infty ue^{-u}du = \frac{1}{n} \to 0 = \int_0^1 f$.

(d) Suppose not. Then there are $\epsilon > 0$ and $x_n, y_n \in K$ with $|x_n - y_n| < 1/n$ but $|g(x_n) - g(y_n)| \geq \epsilon$. By compactness (Bolzano–Weierstrass), a subsequence $x_{n_k} \to x \in K$, and then $y_{n_k} \to x$ as well. By continuity $g(x_{n_k}) - g(y_{n_k}) \to g(x) - g(x) = 0$, contradicting $\geq \epsilon$.

Solution 2 *Analysis: limits and series*

(a) Problem 1(a) with x/n gives $\frac{nx}{n+x} < n\log(1+x/n) < x$. Both bounds tend to x, and exp is continuous.

(b) $|x^k/k!| \leq a^k/k!$ on $[-a,a]$, and $\sum a^k/k! < \infty$ by the ratio test; apply the Weierstrass M-test.

(c) The series of derivatives $\sum_{k\geq 1} x^{k-1}/(k-1)!$ is the same series, uniformly convergent on $[-a,a]$, and the original converges at 0. By the theorem on term-by-term differentiation, $E'(x) = \sum_{k\geq 1} x^{k-1}/(k-1)! = E(x)$.

(d) By the MVT, $f'' \geq 0$ makes f' nondecreasing. For $x < y < z$, the MVT gives $\frac{f(y)-f(x)}{y-x} = f'(\xi_1) \leq f'(\xi_2) = \frac{f(z)-f(y)}{z-y}$, with $\xi_1 < y < \xi_2$. With $y = tx + (1-t)z$, this rearranges to $f(y) \leq tf(x) + (1-t)f(z)$.

Solution 3 *Mean value theorem* **[Past exam 2025-10 #3]**

(a) If f is continuous on $[a,b]$ and differentiable on (a,b), there is $c \in (a,b)$ with $f'(c) = \frac{f(b)-f(a)}{b-a}$. Proof: $g(x) = f(x) - f(a) - \frac{f(b)-f(a)}{b-a}(x-a)$ has $g(a) = g(b) = 0$. By the extreme value theorem g attains a maximum and a minimum on $[a,b]$; if both are at endpoints, $g \equiv 0$; otherwise an interior extremum c has $g'(c) = 0$ (Rolle). (b) $f(x) = x^2$ on $[0,1]$: the chord slope is $1 = f'(\frac{1}{2})$. (c) $f(x) = |x|$ on $[-1,1]$: the chord slope is 0, but $f' = \pm 1$ wherever it exists. Differentiability fails at 0.

Solution 4 *Uniform convergence* **[Past exam 2026-04 #3]**

(a) $f_n(0) = f_n(1) = 0$, and for $0 < x < 1$, $n(1-x)^n \to 0$; so the limit is 0. (b) $f_n'(x) = n(1-x)^{n-1}\{1-(n+1)x\}$, so the maximum is at $x = \frac{1}{n+1}$, with $f_n\big(\frac{1}{n+1}\big) = \big(\frac{n}{n+1}\big)^{n+1} \to e^{-1} \neq 0$. The convergence is not uniform.

Solution 5 *Compactness and completeness* **[Past exam 2024-10 #3]**

Let (a_n) be Cauchy in a compact A. By sequential compactness a subsequence $a_{n_j} \to a \in A$. Given ϵ, choose N with $d(a_n, a_m) < \epsilon/2$ for $n, m \geq N$, and j with $n_j \geq N$ and $d(a_{n_j}, a) < \epsilon/2$. Then $d(a_n, a) < \epsilon$ for $n \geq N$, so $a_n \to a \in A$.

Solution 6 *Linear algebra*

(a) If $P\mathbf{v} = \lambda\mathbf{v}$, $\mathbf{v} \neq \mathbf{0}$, then $\lambda\mathbf{v} = P^2\mathbf{v} = \lambda^2\mathbf{v}$, so $\lambda \in \{0,1\}$. A symmetric P is orthogonally diagonalizable, so its rank is the number of eigenvalues equal to 1, which is their sum, $\operatorname{tr} P$.

(b) $H' = H$, $H^2 = X(X'X)^{-1}X'X(X'X)^{-1}X' = H$, and $\operatorname{tr} H = \operatorname{tr}\{(X'X)^{-1}X'X\} = p$. The same holds for $I - H$, with trace $n - p$, and $(I-H)X = X - X = 0$.

(c) $(I + \mathbf{a}\mathbf{b}')\big(I - \frac{\mathbf{a}\mathbf{b}'}{1+\mathbf{b}'\mathbf{a}}\big) = I + \mathbf{a}\mathbf{b}'\big(1 - \frac{1+\mathbf{b}'\mathbf{a}}{1+\mathbf{b}'\mathbf{a}}\big) = I$. The matrix $\mathbf{a}\mathbf{b}'$ has eigenvalue $\mathbf{b}'\mathbf{a}$ (eigenvector $\mathbf{a}$) and 0 with multiplicity $k-1$, so $\det(I + \mathbf{a}\mathbf{b}') = 1 + \mathbf{b}'\mathbf{a}$.

(d) For $\rho \neq 1$, $\Sigma = (1-\rho)\big(I + \frac{\rho}{1-\rho}\mathbf{1}\mathbf{1}'\big)$, so

$$\Sigma^{-1} = \frac{1}{1-\rho}\Big(I - \frac{\rho}{1+(k-1)\rho}\mathbf{1}\mathbf{1}'\Big).$$

The eigenvalues of Σ are $1+(k-1)\rho$ (eigenvector $\mathbf{1}$) and $1-\rho$ (multiplicity $k-1$), so Σ is positive definite iff $-\frac{1}{k-1} < \rho < 1$.

B. Distributions and transformations

Solution 7 *Joint, marginal and conditional distributions*

(a) The area of D is $\int_0^1 x^2 dx = \frac{1}{3}$, so $f(x,y) = 3$ on D. Then $f_X(x) = \int_0^{x^2} 3\,dy = 3x^2$ on $(0,1)$, and $f_Y(y) = \int_{\sqrt{y}}^1 3\,dx = 3(1-\sqrt{y})$ on $(0,1)$.

(b) Given $X = x$, $Y \sim U(0, x^2)$: $E(Y \mid x) = x^2/2$, $\operatorname{Var}(Y \mid x) = x^4/12$. Then $EY = E(X^2)/2 = \frac{1}{2}\cdot\frac{3}{5} = \frac{3}{10}$, and

$$\operatorname{Var} Y = E\Big(\frac{X^4}{12}\Big) + \operatorname{Var}\Big(\frac{X^2}{2}\Big) = \frac{1}{12}\cdot\frac{3}{7} + \frac{1}{4}\Big(\frac{3}{7} - \frac{9}{25}\Big) = \frac{1}{28} + \frac{3}{175} = \frac{37}{700}.$$

(c) $E(XY) = E\{X\,E(Y \mid X)\} = E(X^3)/2 = \frac{1}{2}\cdot\frac{3}{6} = \frac{1}{4}$, $EX = \frac{3}{4}$, so $\operatorname{Cov}(X,Y) = \frac{1}{4} - \frac{3}{4}\cdot\frac{3}{10} = \frac{1}{40}$. Not independent (nonzero covariance; also the support is not a product set).

(d) $\{y > x/2\} \cap D$ is nonempty only where $x^2 > x/2$, i.e. $x > 1/2$:

$$P(Y > X/2) = \int_{1/2}^1 3\Big(x^2 - \frac{x}{2}\Big)dx = 3\Big[\frac{x^3}{3} - \frac{x^2}{4}\Big]_{1/2}^1 = 3\Big(\frac{1}{12} + \frac{1}{48}\Big) = \frac{5}{16}.$$

Solution 8 *Transformations*

(a) Inverse map $x = uv$, $y = (1-u)v$, $0 < u < 1$, $v > 0$, with $|J| = |\det \begin{pmatrix} v & u \\ -v & 1-u \end{pmatrix}| = v$. So $f(u,v) = e^{-v}v = \mathbf{1}_{(0,1)}(u)\cdot ve^{-v}$: $U \sim U(0,1)$ and $V \sim \text{Gamma}(2,1)$, independent.

(b) $X = UV$ with U independent of V, so $E(X \mid V = v) = vEU = v/2$ and $E(X^2 \mid V = v) = v^2 EU^2 = v^2/3$.

(c) For $w > 0$, $f_W(w) = \int_0^\infty f_X(y+w) f_Y(y)\,dy = \int_0^\infty e^{-(y+w)}e^{-y}dy = \frac{1}{2}e^{-w}$; by symmetry $f_W(w) = \frac{1}{2}e^{-|w|}$, $w \in \mathbb{R}$ (double exponential).

(d) The order statistics have joint pdf $2e^{-y_1-y_2}$ on $0 < y_1 < y_2$. With $m = y_1$, $r = y_2 - y_1$ ($|J| = 1$), $f(m,r) = 2e^{-2m-r} = 2e^{-2m}\cdot e^{-r}$ on $m, r > 0$. So $m \sim \text{Exp}(1/2)$ and $R \sim \text{Exp}(1)$ are independent (memorylessness).

Solution 9 *Uniform distribution on a ball* **[Past exam 2026-04 #2]**

The density is $\frac{3}{4\pi}$ on the ball. (a) Integrating x_3 over a chord of length $2\sqrt{1-x_1^2-x_2^2}$ gives $f(x_1,x_2) = \frac{3}{2\pi}\sqrt{1-x_1^2-x_2^2}$ on the unit disk. (b) The cross-section is a disk of area $\pi(1-x_1^2)$, so $f(x_1) = \frac{3}{4}(1-x_1^2)$ on $[-1,1]$. (c) $f(x_1 \mid x_2 = 0) \propto \sqrt{1-x_1^2}$, i.e. $\frac{2}{\pi}\sqrt{1-x_1^2}$ on $[-1,1]$. (d) By symmetry the means and covariances vanish, and $\operatorname{Var} X_1 = \int_{-1}^1 \frac{3}{4}x^2(1-x^2)dx = \frac{1}{5}$; so $\operatorname{Cov}(\mathbf{X}) = \frac{1}{5}I_3$.

Solution 10 *Conditional expectation and random sums*

(a) $E(S \mid N) = N\mu$ and $\operatorname{Var}(S \mid N) = N\sigma^2$, so $ES = \lambda\mu$ and $\operatorname{Var} S = E(N\sigma^2) + \operatorname{Var}(N\mu) = \lambda(\sigma^2 + \mu^2)$.

(b) $E(e^{tS} \mid N) = M_X(t)^N$, and $Es^N = e^{\lambda(s-1)}$ for Poisson N; so $M_S(t) = \exp\{\lambda(M_X(t) - 1)\}$.

(c) $M_X(t) = 1 - p + pe^t$ gives $M_S(t) = \exp\{\lambda p(e^t - 1)\}$: $S \sim \text{Poisson}(\lambda p)$ (thinning).

(d) Given $N = n$, $(X_1, \ldots, X_n)$ is exchangeable, so $E(X_i \mid S, N = n)$ is the same for all $i \le n$; these sum to $E(S \mid S, N = n) = S$. Hence each equals S/n.

Solution 11 *Poisson process* **[Past exam 2022-04 #3]**

Given two arrivals in an hour, their times are i.i.d. $U(0,60)$ minutes. They overlap iff $|U_1-U_2|<10$, which has probability $1-(50/60)^2=\frac{11}{36}$.

Solution 12 *Minimum of a random number of uniforms* **[Past exam 2023-04 #3]**

$P(Z>z\mid X=x)=(1-z)^x$, so

$$P(Z>z)=c\sum_{x\ge1}\frac{(1-z)^x}{x!}=\frac{e^{1-z}-1}{e-1},\qquad P(Z\le z)=\frac{e-e^{1-z}}{e-1}.$$

Solution 13 *Moment generating functions and cumulants*

(a) $B(5,0.7)$.

(b) $M_Y(t)=M_{X_1}(t)M_{X_2}(2t)=\exp\{2(e^t-1)+(e^{2t}-1)\}$, $EY=2+2\cdot1=4$, $\operatorname{Var}Y=2+4\cdot1=6$. Not Poisson, since its mean and variance differ.

(c) $K(t)=-\alpha\log(1-\beta t)=\alpha\sum_{r\ge1}\frac{(\beta t)^r}{r}$, so $\kappa_r=r!\cdot\frac{\alpha\beta^r}{r}=\alpha\beta^r(r-1)!$. Skewness $=\kappa_3/\kappa_2^{3/2}=2\alpha\beta^3/(\alpha\beta^2)^{3/2}=2/\sqrt{\alpha}$.

(d) $M(t)=\sum_k EX^k\frac{t^k}{k!}=\sum_k(k+1)t^k=(1-t)^{-2}$ for $|t|<1$. This is the Gamma$(2,1)$ mgf, and an mgf finite near 0 determines the distribution.

Solution 14 *Uniform order statistics*

(a) $f(u,v)=n(n-1)(v-u)^{n-2}$ on $0<u<v<1$.

(b) With $r=v-u$ and $u\in(0,1-r)$, $f_R(r)=\int_0^{1-r}n(n-1)r^{n-2}du=n(n-1)r^{n-2}(1-r)$, the Beta$(n-1,2)$ density $(1/B(n-1,2)=n(n-1))$. $ER=\frac{n-1}{n+1}$.

(c) $U_{(1)}\sim$ Beta$(1,n)$ and $U_{(n)}\sim$ Beta$(n,1)$ both have variance $\frac{n}{(n+1)^2(n+2)}$, and $\operatorname{Var}R=\frac{2(n-1)}{(n+1)^2(n+2)}$. From $\operatorname{Var}R=\operatorname{Var}U_{(n)}+\operatorname{Var}U_{(1)}-2\operatorname{Cov}$,

$$\operatorname{Cov}(U_{(1)},U_{(n)})=\frac{2n-2(n-1)}{2(n+1)^2(n+2)}=\frac{1}{(n+1)^2(n+2)}.$$

(d) $1-P(\text{all}<\frac12)-P(\text{all}>\frac12)=1-2^{1-n}$.

Solution 15 *Logistic order statistics* **[Past exam 2022-10 #1]**

With $F(x)=1/(1+e^{-x})$, $1+e^{-x}=1/F(x)$, so the variable equals $\log\{F(X_{(n)})/F(X_{(1)})\}$. Let $V_i=-\log F(X_i)\overset{\text{iid}}{\sim}\text{Exp}(1)$; the map is decreasing, so the variable is $\max V_i-\min V_i$. By memorylessness this range has the law of the maximum of $n-1$ i.i.d. Exp(1) variables:

$$f(y)=(n-1)e^{-y}(1-e^{-y})^{n-2},\qquad y>0.$$

Solution 16 *Multivariate normal*

(a) With $a = (1, -1, 1)'$, $\Sigma a = (1, 0, 1)'$ and $a'\Sigma a = 2$. So $X_1 - X_2 + X_3 \sim N(0, 2)$.

(b) $(X_2 - c(X_1 + X_3),\ X_1 + X_3)$ is a linear image of $\mathbf{X}$, hence bivariate normal, and is independent iff the covariance is 0. $\mathrm{Cov}(X_2, X_1 + X_3) = 1 + 1 = 2$ and $\mathrm{Var}(X_1 + X_3) = 2 + 2 + 2 \cdot 0 = 4$, so $\mathrm{Cov} = 2 - 4c = 0$ gives $c = \frac{1}{2}$.

(c) $\Sigma_{12} = (1, 1)$, $\Sigma_{22} = 2I_2$, so $\Sigma_{12}\Sigma_{22}^{-1} = (\frac{1}{2}, \frac{1}{2})$. Hence $X_2 \mid (x_1, x_3) \sim N\big(\frac{x_1+x_3}{2},\ 2 - \frac{1}{2}(1+1)\big) = N\big(\frac{x_1+x_3}{2}, 1\big)$.

(d) $\mathrm{Cov}(X_1, X_3) = 0$ and they are jointly normal, so $X_1, X_3 \overset{\text{iid}}{\sim} N(0, 2)$. Conditioning on X_3, $E(e^{tX_1X_3} \mid X_3) = \exp(\frac{1}{2} \cdot 2t^2X_3^2) = e^{t^2X_3^2}$. Since $X_3^2/2 \sim \chi^2(1)$, $Ee^{t^2X_3^2} = Ee^{2t^2(X_3^2/2)} = (1 - 4t^2)^{-1/2}$. So $M(t) = (1 - 4t^2)^{-1/2}$ for $|t| < \frac{1}{2}$, and $M(t) = \infty$ otherwise.

(e) Σ is positive definite (leading minors $2, 3, 4$). $\mathbf{Z} = \Sigma^{-1/2}\mathbf{X} \sim N_3(\mathbf{0}, I)$ and $\mathbf{X}'\Sigma^{-1}\mathbf{X} = \mathbf{Z}'\mathbf{Z} \sim \chi^2(3)$.

Solution 17 *Bivariate normal*

(a) Write $Y = \rho X + \sqrt{1 - \rho^2}Z$ with X, Z i.i.d. $N(0, 1)$, and $X = R\cos\Theta$, $Z = R\sin\Theta$ with $\Theta \sim U(-\pi, \pi)$ independent of R. With $\phi = \arcsin\rho$, $Y = R\sin(\Theta + \phi)$. So $\{X > 0, Y > 0\} = \{\Theta \in (-\frac{\pi}{2}, \frac{\pi}{2}) \cap (-\phi, \pi - \phi)\} = \{\Theta \in (-\phi, \frac{\pi}{2})\}$, of probability $\frac{\pi/2+\phi}{2\pi} = \frac{1}{4} + \frac{\arcsin\rho}{2\pi}$.

(b) $\max(X, Y) = \frac{X+Y}{2} + \frac{|X-Y|}{2}$, and $X - Y \sim N(0, 2(1 - \rho))$ with $E|X - Y| = \sqrt{2(1-\rho)}\sqrt{2/\pi}$. Hence $E\max(X, Y) = \sqrt{(1 - \rho)/\pi}$.

(c) $Y \mid X = x \sim N(\rho x, 1 - \rho^2)$, so $E(Y^2 \mid x) = \rho^2x^2 + 1 - \rho^2$. Then $EX^2Y^2 = E\{X^2(\rho^2X^2 + 1 - \rho^2)\} = 3\rho^2 + 1 - \rho^2$ and $\mathrm{Cov}(X^2, Y^2) = 2\rho^2$.

Solution 18 *Multivariate normal with AR(1) covariance* **[Past exam 2025-10 #2]**

(a) $m(\mathbf{t}) = \exp\{\frac{1}{2}(t_1^2 + t_2^2 + t_3^2) + \rho t_1t_2 + \rho t_2t_3 + \rho^2t_1t_3\}$. (b) With $\mathbf{a} = (1, -2, 1)'$, $\Sigma\mathbf{a} = ((1 - \rho)^2,\ -2(1-\rho),\ (1-\rho)^2)'$, so $\mathbf{a}'\Sigma\mathbf{a} = 2(1-\rho)^2 + 4(1-\rho) = 2(1-\rho)(3-\rho)$. (c) $\mathrm{Cov}(X_1 - X_2, X_2 - X_3) = \rho - \rho^2 - 1 + \rho = -(1-\rho)^2 \neq 0$. They are jointly normal with nonzero covariance, hence dependent.

Solution 19 *Normal samples and prediction*

(a) $X_{n+1} - \bar{X} \sim N(0, \sigma^2(1 + 1/n))$, and it is independent of S^2 (X_{n+1} is independent of the first n, and $\bar{X}$ is independent of S^2). Dividing by $S\sqrt{1 + 1/n}$ gives $Z/\sqrt{V/(n-1)}$ with $V = (n-1)S^2/\sigma^2 \sim \chi^2(n-1)$: a $t(n-1)$ variable. Prediction interval: $\bar{X} \pm t_{0.025}(n-1)S\sqrt{1 + 1/n}$.

(b) With $V \sim \chi^2(n-1)$,

$$E\sqrt{V} = \int_0^\infty \frac{v^{n/2-1}e^{-v/2}}{\Gamma(\frac{n-1}{2})2^{(n-1)/2}}dv = \frac{\Gamma(\frac{n}{2})2^{n/2}}{\Gamma(\frac{n-1}{2})2^{(n-1)/2}} = \sqrt{2}\frac{\Gamma(n/2)}{\Gamma((n-1)/2)},$$

and $ES = \frac{\sigma}{\sqrt{n-1}}E\sqrt{V}$.

(c) With $\sum(X_i - \bar{X})^2 = \sigma^2V$, $EV = n - 1$, $EV^2 = 2(n-1) + (n-1)^2$,

$$E(c\sigma^2V - \sigma^2)^2 = \sigma^4\{c^2(n-1)(n+1) - 2c(n-1) + 1\},$$

minimized at $c = \frac{1}{n+1}$.

Solution 20 *Chi-square, t and F*

(a) $T^2 = Z^2/(V/r)$ with $Z^2 \sim \chi^2(1)$ independent of $V \sim \chi^2(r)$: this is $F(1, r)$.

(b) Write $X = \frac{V_1/r_1}{V_2/r_2}$. Then $\frac{r_1 X}{r_2 + r_1 X} = \frac{V_1}{V_1+V_2}$ with independent $V_i \sim \text{Gamma}(r_i/2, 2)$, which is $\text{Beta}(r_1/2, r_2/2)$.

(c) $2n\bar{X}/\theta \sim \chi^2(2n)$ and $2m\bar{Y}/\theta \sim \chi^2(2m)$ are independent, and $\bar{X}/\bar{Y} = \frac{(2n\bar{X}/\theta)/(2n)}{(2m\bar{Y}/\theta)/(2m)} \sim F(2n, 2m)$.

(d) $EX = \frac{r_2}{r_1} EV_1\, EV_2^{-1} = \frac{r_2}{r_1} \cdot r_1 \cdot \frac{1}{r_2 - 2}$, since $EV^{-1} = \frac{\Gamma(r/2-1)2^{r/2-1}}{\Gamma(r/2)2^{r/2}} = \frac{1}{r-2}$ for $V \sim \chi^2(r)$.

Solution 21 *Quadratic forms*

(a) $\mathbf{a} = (1, -1, 0)'/\sqrt{2}$ and $\mathbf{b} = (1, 1, -2)'/\sqrt{6}$ are orthonormal, so $U = \mathbf{a}'\mathbf{Z}$, $V = \mathbf{b}'\mathbf{Z}$ are i.i.d. $N(0, 1)$ and $Q = U^2 + V^2 \sim \chi^2(2)$. Together with $\mathbf{1}/\sqrt{3}$ they form an orthonormal basis, so $\sum Z_i^2 = 3\bar{Z}^2 + U^2 + V^2$, i.e. $Q = \sum Z_i^2 - 3\bar{Z}^2 = \sum (Z_i - \bar{Z})^2$.

(b) Write $A = P\Lambda P'$ with P orthogonal and $\mathbf{Y} = P'\mathbf{Z} \sim N(\mathbf{0}, I)$. Then $\mathbf{Z}'A\mathbf{Z} = \sum \lambda_i Y_i^2$, with mean $\sum \lambda_i = \operatorname{tr} A$ and variance $\sum \lambda_i^2 \operatorname{Var}(Y_i^2) = 2\sum \lambda_i^2 = 2 \operatorname{tr} A^2$.

(c) J and $I - J$ are symmetric idempotent and $J(I - J) = 0$, so $J\mathbf{Z}$ and $(I - J)\mathbf{Z}$ are uncorrelated jointly normal vectors, hence independent; the quadratic forms are their squared lengths. Since $\operatorname{tr} J = 1$ and $\operatorname{tr}(I - J) = n - 1$, they are $\chi^2(1)$ and $\chi^2(n - 1)$.

Solution 22 *Independence of quadratic forms* **[Past exam 2022-10 #5]**

Sufficiency. $A\mathbf{y}$ and $B\mathbf{y}$ are jointly normal with $\operatorname{Cov}(B\mathbf{y}, A\mathbf{y}) = BVA$. If $BVA = 0$ (equivalently $AVB = 0$), they are independent. Then $\mathbf{y}'A\mathbf{y} = \|A\mathbf{y}\|^2$ is a function of $A\mathbf{y}$, which gives both (1) and (2).

Necessity (sketch, $\mu = 0$). Write $\mathbf{y} = V^{1/2}\mathbf{z}$. In (1), independence gives $\operatorname{Cov}(\mathbf{y}'A\mathbf{y}, (\mathbf{b}'B\mathbf{y})^2) = 0$ for every $\mathbf{b}$, and this covariance equals $2\,\mathbf{b}'BVAVB\mathbf{b} = 2\|AVB\mathbf{b}\|^2$ (using $A = A^2$). Hence $AVB = 0$. In (2), $\operatorname{Cov}(\mathbf{y}'A\mathbf{y}, \mathbf{y}'B\mathbf{y}) = 2\operatorname{tr}(AVBV) = 2\operatorname{tr}(PQ)$ with $P = V^{1/2}AV^{1/2}$ and $Q = V^{1/2}BV^{1/2}$ nonnegative definite. $\operatorname{tr}(PQ) = \|P^{1/2}Q^{1/2}\|_F^2 = 0$ forces $PQ = 0$, i.e. $AVB = 0$. For general μ the same conclusion follows from the cumulants.

Solution 23 *Distributions of quadratic forms* **[Past exam 2022-10 #6]**

H and $I - H$ are symmetric idempotent with ranks k and $n - k$, and $H(I - H) = 0$. So $\mathbf{y}'H\mathbf{y} \sim \chi^2(k)$ and $\mathbf{y}'(I - H)\mathbf{y} \sim \chi^2(n - k)$, independently. Hence $\frac{n-k}{k} T \sim F(k, n - k)$, and $T/(1 + T) \sim \text{Beta}(\frac{k}{2}, \frac{n-k}{2})$.

Solution 24 *Expectation of a quadratic form* **[Past exam 2023-04 #1]**

$\mathbf{y}'A\mathbf{y} = \operatorname{tr}(A\mathbf{y}\mathbf{y}')$ and $E\mathbf{y}\mathbf{y}' = \Sigma + \mu\mu'$, so $E\mathbf{y}'A\mathbf{y} = \operatorname{tr}(A\Sigma) + \operatorname{tr}(A\mu\mu') = \operatorname{tr}(A\Sigma) + \mu' A\mu$.

Solution 25 *Distribution of an F statistic* **[Past exam 2024-05 #4]**

$I - H$ annihilates $X\beta$, so $\mathbf{y}'(I - H)\mathbf{y}/\sigma^2 \sim \chi^2(n - p - 1)$, while $\mathbf{y}'H\mathbf{y}/\sigma^2 \sim \chi^2(p + 1; \delta)$ with $\delta = \beta' X'X\beta/\sigma^2$ (Theorem 10.11 of the notes). They are independent since $H(I - H) = 0$. Hence $\frac{n-p-1}{p+1} F \sim F(p + 1, n - p - 1; \delta)$, a central F only when $\beta = \mathbf{0}$.

C. Convergence and limit theorems

Solution 26 *Stochastic order and limits*

(a) Let F be the cdf of X and $\epsilon > 0$. Since F has at most countably many discontinuities, choose M such that $\pm M$ are continuity points and $P(X > M) + P(X \le -M) < \epsilon/2$. Then

$$P(|X_n| > M) \le 1 - F_n(M) + F_n(-M) \to 1 - F(M) + F(-M) < \epsilon/2,$$

so $P(|X_n| > M) < \epsilon$ for all large n; the finitely many earlier n are handled by enlarging M.

(b) For $\delta, \epsilon > 0$, choose M with $P(|X_n| > M) < \epsilon$ for large n. Then

$$P(|X_n Y_n| > \delta) \le P(|X_n| > M) + P(|Y_n| > \delta/M) < \epsilon + o(1).$$

(c) For $0 \le x \le n\theta$, $P(n(\theta - X_{(n)}) > x) = P(X_{(n)} < \theta - x/n) = (1 - \frac{x}{n\theta})^n \to e^{-x/\theta}$. So $n(\theta - X_{(n)}) \xrightarrow{d} \mathrm{Exp}(\theta)$, which is $O_p(1)$ by (a), i.e. $X_{(n)} - \theta = O_p(n^{-1})$. Then $\sqrt{n}(X_{(n)} - \theta) = n^{-1/2} \cdot n(X_{(n)} - \theta) = o(1)O_p(1) = o_p(1)$ by (b).

(d) By the CLT, $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$, so $2\mu\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, 4\mu^2\sigma^2)$. Also $\sqrt{n}(\bar{X}_n - \mu)^2 = \sqrt{n}(\bar{X}_n - \mu) \cdot (\bar{X}_n - \mu) = O_p(1)\, o_p(1) = o_p(1)$, by (a), the WLLN and (b). Slutsky's theorem gives the result.

Solution 27 *Tightness* **[Past exam 2024-10 #2]**

See Problem 26(a). Since the cdf of Z is continuous, every $\pm k$ is a continuity point, and

$$\limsup_n P(|X_n| > k) = P(|Z| > k) \to 0 \qquad (k \to \infty).$$

The finitely many early n are handled by enlarging k.

Solution 28 *O_p and o_p algebra* **[Past exam 2025-04 #1]**

(1) $P(|X_n + Y_n| > 2k) \le P(|X_n| > k) + P(|Y_n| > k)$. (2) $P(|X_n Y_n| > k^2) \le P(|X_n| > k) + P(|Y_n| > k)$. Take $\sup_n$ and let $k \to \infty$ in both. (3) See Problem 26(b). (4) Given ϵ, choose N with $P(|Z_n| > 1) < \epsilon$ for $n \ge N$. For each of the finitely many $n < N$, $P(|Z_n| > k) \to 0$ as $k \to \infty$. So $\sup_n P(|Z_n| > k) < \epsilon$ for all large k.

Solution 29 *Modes of convergence*

(a) $P(|X_n| > \epsilon) = 1/n \to 0$, but $EX_n = n \cdot \frac{1}{n} = 1$ for all n.

(b) By symmetry, $X_n \sim N(0,1)$ for every n, so $X_n \xrightarrow{d} Z$ trivially. For odd n, $X_n - Z = -2Z$ and $P(|2Z| > \epsilon)$ is a fixed positive number, so $X_n \not\xrightarrow{p} Z$.

(c) For $\epsilon > 0$, $c - \epsilon$ and $c + \epsilon/2$ are continuity points of the limit cdf $\mathbf{1}\{x \ge c\}$, so

$$P(|X_n - c| \ge \epsilon) \le P(X_n \le c - \epsilon) + 1 - P(X_n \le c + \epsilon/2) \to 0 + 0.$$

(d) Markov's inequality: $P(|X_n - a| \ge \epsilon) \le E(X_n - a)^2/\epsilon^2 \to 0$.

Solution 30 *Probability inequalities*

(a) For $0 < r < s$, $u \mapsto u^{s/r}$ is convex on $[0, \infty)$, so by Jensen $E|X|^s = E\{(|X|^r)^{s/r}\} \geq (E|X|^r)^{s/r}$. Take the $1/s$ power.

(b) $\operatorname{Var} \hat{p} = p(1-p)/n \leq 1/(4n)$, so $P(|\hat{p} - p| \geq 0.05) \leq \frac{1}{4n(0.05)^2} = \frac{100}{n} \leq 0.05$ once $n \geq 2000$.

(c) For $t > 0$, Markov's inequality applied to e^{tX} gives $P(X \geq 2\lambda) \leq e^{-2\lambda t} E e^{tX} = \exp\{-2\lambda t + \lambda(e^t - 1)\}$. The exponent is minimized at $e^t = 2$, giving $P(X \geq 2\lambda) \leq e^{\lambda(1 - 2\log 2)} = (e/4)^{\lambda}$.

Solution 31 *Correlation and convergence in distribution* **[Past exam 2022-10 #2]**

Let A_n, B_n be the standardized X_n, Y_n. Then $E(A_n - B_n)^2 = 2 - 2\operatorname{Corr}(X_n, Y_n) \to 0$, so $B_n - A_n \xrightarrow{p} 0$ by Markov's inequality. Since $B_n = A_n + (B_n - A_n)$, Slutsky's theorem gives $B_n \xrightarrow{d} X$.

Solution 32 *Central limit theorem in practice*

(a) The sum has mean 24 and variance $48/12 = 4$, so $P \approx 1 - \Phi\big(\frac{26-24}{2}\big) = 1 - \Phi(1) = 0.159$.

(b) Since $p(1-p) \leq \frac{1}{4}$, we need $1.96\sqrt{0.25/n} \leq 0.03$, i.e. $n \geq (1.96/0.06)^2 = 1067.1$, so $n = 1068$.

(c) $P(X \geq 120) = P(X > 119.5) \approx 1 - \Phi\big(\frac{19.5}{10}\big) = 1 - \Phi(1.95) = 0.026$.

Solution 33 *Delta method and variance stabilization*

(a) $g(\theta) = \sqrt{\theta}$, $g'(\theta) = \frac{1}{2\sqrt{\theta}}$, and $\sqrt{n}(\bar{X} - \theta) \xrightarrow{d} N(0, \theta)$, so $\sqrt{n}(\sqrt{\bar{X}} - \sqrt{\theta}) \xrightarrow{d} N(0, \frac{1}{4})$, free of θ. Hence $\sqrt{\theta} \in \sqrt{\bar{X}} \pm \frac{1.96}{2\sqrt{n}}$, and squaring (lower end truncated at 0) gives the interval for θ.

(b) $g = \log$, $g'(\theta) = 1/\theta$, variance $\theta^2 \cdot \theta^{-2} = 1$: the limit is $N(0, 1)$.

(c) $g(p) = \arcsin\sqrt{p}$ has $g'(p) = \frac{1}{2\sqrt{p(1-p)}}$, so the limit is $N\big(0, \frac{p(1-p)}{4p(1-p)}\big) = N(0, \frac{1}{4})$.

Solution 34 *Asymptotics of the sample standard deviation* **[Past exam 2024-05 #1]**

$(n-1)S_n^2/\sigma^2$ is distributed as a sum of $n-1$ i.i.d. $\chi^2(1)$ variables (mean 1, variance 2), so by the CLT $\sqrt{n-1}(S_n^2/\sigma^2 - 1) \xrightarrow{d} N(0, 2)$. The delta method with $g(u) = \sqrt{u}$, $g'(1) = \frac{1}{2}$, gives $\sqrt{n-1}(S_n/\sigma - 1) \xrightarrow{d} N(0, \frac{1}{2})$; multiply by σ.

Solution 35 *Fast convergence of S_n* **[Past exam 2024-05 #2]**

Since $X_i^2 = \sigma^2$, $(n-1)S_n^2 = \sum X_i^2 - n\bar{X}^2 = n\sigma^2 - n\bar{X}^2$, so $S_n^2 - \sigma^2 = \frac{\sigma^2 - n\bar{X}^2}{n-1}$. By the CLT, $n\bar{X}^2 = (\sqrt{n}\bar{X})^2 = O_p(1)$, so $S_n^2 - \sigma^2 = O_p(n^{-1})$. Then $|S_n - \sigma| = |S_n^2 - \sigma^2|/(S_n + \sigma) \leq |S_n^2 - \sigma^2|/\sigma = O_p(n^{-1})$, which is $o_p((n-1)^{-\gamma})$ for every $\gamma < 1$. The rate is faster than $n^{-1/2}$ because $\mu_4 = \sigma^4$ makes the limit variance $(\mu_4 - \sigma^4)/(4\sigma^2)$ vanish.

Solution 36 *Extremes*

(a) For $x > -\log n$, $P(X_{(n)} \le \log n + x) = (1 - e^{-x}/n)^n \to \exp(-e^{-x})$.

(b) $P(nX_{(1)} > s) = (e^{-s/n})^n = e^{-s}$: Exp(1) exactly.

(c) For $s > 0$ and $x > -\log n$,

$$P\big(nX_{(1)} > s,\ X_{(n)} - \log n \le x\big) = P\big(s/n < X_i \le \log n + x\ \forall i\big) = \Big(e^{-s/n} - \frac{e^{-x}}{n}\Big)^n \to e^{-s}\exp(-e^{-x}),$$

since $n\{e^{-s/n} - 1 - e^{-x}/n\} \to -s - e^{-x}$. The joint limit cdf factors, so the limits are independent.

Solution 37 *Asymptotic independence of extremes* **[Past exam 2021-10 #5]**

True. With $U_i = \Phi(X_i) \overset{\text{iid}}{\sim} U(0,1)$, the two variables are $nU_{(1)}$ and $n(1 - U_{(n)})$. For $s, t \ge 0$ and $s + t < n$,

$$P\big(nU_{(1)} > s,\ n(1 - U_{(n)}) > t\big) = P\Big(\frac{s}{n} < U_i < 1 - \frac{t}{n}\ \forall i\Big) = \Big(1 - \frac{s+t}{n}\Big)^n \to e^{-s}e^{-t}.$$

The joint survival function converges to that of two independent Exp(1) variables, so they are asymptotically independent.

Solution 38 *Limit of an order-statistic functional* **[Past exam 2020-10 #1]**

Since $\frac{d}{dx}\{-\log(1 - \Phi(x))\} = \frac{\phi(x)}{1-\Phi(x)}$, put $E_i = -\log(1 - \Phi(X_i))$. These are i.i.d. Exp(1) (because $1 - \Phi(X_i) \sim U(0,1)$), and the map is increasing, so $E_{(j)} = -\log(1 - \Phi(X_{(j)}))$ with $E_{(0)} = 0$. The integral is $E_{(n-i+1)} - E_{(n-i)}$. With $j = n - i + 1$, the term is $(n - j + 1)(E_{(j)} - E_{(j-1)}) =: D_j$, and the normalized spacings $D_1, \dots, D_n$ are i.i.d. Exp(1).

(a) $T_n = \sum_{j=1}^{n-k} D_j \sim \text{Gamma}(n - k, 1)$.

(b) For $n = 2k$, $T_n \sim \text{Gamma}(k, 1)$. Take $a_k = k^{-1/2}$, $b_k = k$. The cgf of $(T_n - k)/\sqrt{k}$ is

$$-t\sqrt{k} - k\log\Big(1 - \frac{t}{\sqrt{k}}\Big) = -t\sqrt{k} + k\Big(\frac{t}{\sqrt{k}} + \frac{t^2}{2k} + O(k^{-3/2})\Big) = \frac{t^2}{2} + O(k^{-1/2})$$

for $|t| < \sqrt{k}$, so the mgf converges to $e^{t^2/2}$.

D. Point estimation

Solution 39 *MLE, MME and efficiency*

(a) $P(X^2 > t) = \int_{\sqrt{t}}^{\infty} \frac{x}{\theta} e^{-x^2/2\theta} dx = e^{-t/(2\theta)}$, so $X^2 \sim \text{Exp}(2\theta)$. The log-likelihood is $l(\theta) = \sum \log x_i - n \log \theta - \sum x_i^2/(2\theta)$, with $\dot{l} = -n/\theta + \sum x_i^2/(2\theta^2)$ positive iff $\theta < \sum x_i^2/(2n)$. So $\hat{\theta} = \frac{1}{2n} \sum X_i^2$, with $E\hat{\theta} = \theta$ and $\operatorname{Var} \hat{\theta} = \frac{(2\theta)^2}{4n} = \frac{\theta^2}{n}$.

(b) $\partial_\theta^2 \log f = \frac{1}{\theta^2} - \frac{x^2}{\theta^3}$, so $I(\theta) = -\frac{1}{\theta^2} + \frac{2\theta}{\theta^3} = \frac{1}{\theta^2}$. The CR bound for unbiased estimators is $\frac{1}{nI(\theta)} = \frac{\theta^2}{n} = \operatorname{Var} \hat{\theta}$. So $\hat{\theta}$ is efficient, hence UMVUE (also: $\sum X_i^2$ is complete sufficient in this exponential family).

(c) $\theta = \frac{2}{\pi}(EX)^2$, so $\tilde{\theta} = \frac{2}{\pi}\bar{X}^2$. $\operatorname{Var} X = EX^2 - (EX)^2 = 2\theta - \frac{\pi\theta}{2} = \frac{(4-\pi)\theta}{2}$. With $g(m) = 2m^2/\pi$, $g'(m) = 4m/\pi$ and $m = \sqrt{\pi\theta/2}$, the delta method gives

$$\sqrt{n}(\tilde{\theta} - \theta) \xrightarrow{d} N\Big(0, \frac{16m^2}{\pi^2} \cdot \frac{(4-\pi)\theta}{2}\Big) = N\Big(0, \frac{4(4-\pi)}{\pi}\theta^2\Big).$$

(d) $ARE(\tilde{\theta}, \hat{\theta}) = \dfrac{\theta^2}{4(4-\pi)\theta^2/\pi} = \dfrac{\pi}{4(4-\pi)} \approx 0.915.$

Solution 40 *Multinomial reparametrization*

(a) $l(\theta) = (2N_1 + N_2)\log\theta + (N_2 + 2N_3)\log(1-\theta) + \text{const}$, strictly concave, so $\hat{\theta} = \frac{2N_1+N_2}{2n}$.

(b) Each individual contributes $Y_i \in \{2, 1, 0\}$ copies of the allele, with probabilities $\theta^2, 2\theta(1-\theta), (1-\theta)^2$, i.e. $Y_i \sim B(2, \theta)$, independently. So $2N_1 + N_2 = \sum Y_i \sim B(2n, \theta)$, $E\hat{\theta} = \theta$, $\operatorname{Var} \hat{\theta} = \frac{\theta(1-\theta)}{2n}$.

(c) Per individual, $I(\theta) = \operatorname{Var}\left(\frac{Y}{\theta} - \frac{2-Y}{1-\theta}\right) = \frac{2\theta(1-\theta)}{\theta^2(1-\theta)^2} = \frac{2}{\theta(1-\theta)}$. Hence $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \theta(1-\theta)/2)$ (here even $\operatorname{Var} \hat{\theta} = 1/(nI)$ exactly). $2N_1 + N_2$ is complete sufficient (binomial family), and $\hat{\theta}$ is an unbiased function of it: UMVUE.

(d) $\hat{\theta} = (100 + 90)/400 = 0.475$. Expected counts: $200(0.475)^2 = 45.125$, $200 \cdot 2(0.475)(0.525) = 99.75$, $200(0.525)^2 = 55.125$. Pearson's statistic:

$$\frac{4.875^2}{45.125} + \frac{9.75^2}{99.75} + \frac{4.875^2}{55.125} = 0.527 + 0.953 + 0.431 = 1.91.$$

Degrees of freedom $= \dim \Omega - \dim \Omega_0 = 2 - 1 = 1$; since $1.91 < \chi^2_{0.05}(1) = 3.84$, the model is not rejected.

Solution 41 *Multinomial reparametrization* **[Past exam 2026-04 #1]**

(a) The likelihood is $\theta_1^{x_1}(1-\theta_1)^{x_2+x_3}\theta_2^{x_2}(1-\theta_2)^{x_3}$, which separates, so $\hat{\theta}_1 = X_1/n$ and $\hat{\theta}_2 = X_2/(X_2+X_3)$. (b) For the n trials, $I = \operatorname{diag}\Big(\frac{n}{\theta_1(1-\theta_1)}, \frac{n(1-\theta_1)}{\theta_2(1-\theta_2)}\Big)$, using $E(X_2+X_3) = n(1-\theta_1)$; the off-diagonal entries vanish because the log-likelihood is additive. (c) $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N_2\Big(\mathbf{0}, \operatorname{diag}\big(\theta_1(1-\theta_1), \frac{\theta_2(1-\theta_2)}{1-\theta_1}\big)\Big)$: the two estimators are asymptotically independent.

Solution 42 *Beta model: MME and MLE* **[Past exam 2025-04 #2]**

(1) $m_1 = \frac{\alpha}{\alpha+1}$ gives $\alpha = \frac{m_1}{1-m_1}$, hence the MME. With $g(m) = \frac{m}{1-m}$, $g'(m_1) = (\alpha+1)^2$, and $\operatorname{Var} X = \frac{\alpha}{(\alpha+1)^2(\alpha+2)}$, so $\sqrt{n}(\hat{\alpha}^{\mathrm{MME}} - \alpha) \xrightarrow{d} N\big(0, \frac{\alpha(\alpha+1)^2}{\alpha+2}\big)$. (2) $\hat{\alpha}^{\mathrm{MLE}} = -n/\sum \log X_i$; $I(\alpha) = \alpha^{-2}$, so the limit is $N(0, \alpha^2)$. (3) $ARE = \frac{\alpha(\alpha+2)}{(\alpha+1)^2} < 1$.

Solution 43 *MLE and MME compared by MSE* **[Past exam 2025-10 #1]**

(a) $L = 2^n \prod x_i \, \theta^{-2n} \mathbf{1}\{\theta \geq x_{(n)}\}$ is decreasing in θ, so $\hat{\theta}^{\mathrm{MLE}} = X_{(n)}$. $EX = \frac{2}{3}\theta$ gives $\hat{\theta}^{\mathrm{MME}} = \frac{3}{2}\bar{X}$.

(b) $X_{(n)}$ has cdf $(x/\theta)^{2n}$: $EX_{(n)} = \frac{2n}{2n+1}\theta$, $EX_{(n)}^2 = \frac{n}{n+1}\theta^2$, so $\operatorname{Var} X_{(n)} = \frac{n\theta^2}{(n+1)(2n+1)^2}$. $\hat{\theta}^{\mathrm{MME}}$ is unbiased, and with $\operatorname{Var} X = \frac{\theta^2}{2} - \frac{4\theta^2}{9} = \frac{\theta^2}{18}$, $\operatorname{Var} \hat{\theta}^{\mathrm{MME}} = \frac{9}{4} \cdot \frac{\theta^2}{18n} = \frac{\theta^2}{8n}$.

(c) $MSE(\hat{\theta}^{\mathrm{MLE}}) = \frac{n\theta^2}{(n+1)(2n+1)^2} + \frac{\theta^2}{(2n+1)^2} = \frac{\theta^2}{(n+1)(2n+1)}$. This is below $\frac{\theta^2}{8n}$ iff $2n^2 - 5n + 1 > 0$, i.e. $n \geq 3$. For $n = 1, 2$ the MME has smaller MSE; for $n \geq 3$ the MLE does, and its MSE decays like n^{-2}.

Solution 44 *MME and MLE for a uniform model* **[Past exam 2023-10 #3]**

For $\theta > 0$ the support is $(\theta, 2\theta)$, and for $\theta < 0$ it is $(\theta, 0)$; the sign of the data reveals the sign of θ. (1) $EX = \frac{3}{2}\theta$ for $\theta > 0$ and $\frac{1}{2}\theta$ for $\theta < 0$, so $\tilde{\theta} = \frac{2}{3}\bar{X}$ if $\bar{X} > 0$ and $2\bar{X}$ if $\bar{X} < 0$. (2) For positive data, $L = \theta^{-n}\mathbf{1}\{x_{(n)}/2 \leq \theta \leq x_{(1)}\}$ gives $\hat{\theta} = X_{(n)}/2$. For negative data, $L = |\theta|^{-n}\mathbf{1}\{\theta \leq x_{(1)}\}$ gives $\hat{\theta} = X_{(1)}$ (with closed support). (3) Yes: $X_{(n)} \xrightarrow{p} 2\theta$ for $\theta > 0$ and $X_{(1)} \xrightarrow{p} \theta$ for $\theta < 0$.

Solution 45 *Cauchy location*

(a) $Ee^{it\bar{X}} = \{\varphi(t/n)\}^n = e^{i\theta t - |t|}$, so $\bar{X} \overset{d}{=} X_1$. Then $P(|\bar{X} - \theta| > \epsilon) = P(|X_1 - \theta| > \epsilon)$ does not tend to 0: $\bar{X}$ is not consistent.

(b) The median of X_1 is θ with $f(\theta) = 1/\pi$, so $\sqrt{n}(\hat{m}_n - \theta) \xrightarrow{d} N(0, \frac{1}{4f(\theta)^2}) = N(0, \frac{\pi^2}{4})$.

(c) The score is $\frac{2(x-\theta)}{1+(x-\theta)^2}$, so with Y standard Cauchy,

$$I(\theta) = E\frac{4Y^2}{(1+Y^2)^2} = \frac{4}{\pi}\int_{-\infty}^{\infty} \frac{y^2}{(1+y^2)^3} dy = \frac{4}{\pi} \cdot \frac{\pi}{8} = \frac{1}{2},$$

using $y = \tan u$: $\int \sin^2 u \cos^2 u \, du = \frac{\pi}{8}$ over $(-\frac{\pi}{2}, \frac{\pi}{2})$. The efficient limiting variance is $1/I = 2$, so $ARE(\text{median}, \text{MLE}) = \frac{2}{\pi^2/4} = \frac{8}{\pi^2} \approx 0.81$.

Solution 46 *A non-regular scale model*

(a) $L(\theta) = \theta^{-n}\mathbf{1}\{x_{(n)}/2 \leq \theta \leq x_{(1)}\}$, decreasing on its support, so $\hat{\theta} = X_{(n)}/2$. L depends on the data only through $(x_{(1)}, x_{(n)})$, and two samples give proportional likelihoods only if these agree, so $(X_{(1)}, X_{(n)})$ is minimal sufficient.

(b) $X_i = \theta(1 + U_i)$ with $U_i \overset{\text{iid}}{\sim} U(0,1)$, so $EX_{(n)} = \theta\frac{2n+1}{n+1}$ and $EX_{(1)} = \theta\frac{n+2}{n+1}$. Thus $\frac{n+1}{2n+1}X_{(n)} = \frac{2(n+1)}{2n+1}\hat{\theta}$ is unbiased.

(c) $g = (2n+1)X_{(1)} - (n+2)X_{(n)}$ has $E_\theta g = \frac{\theta}{n+1}\{(2n+1)(n+2) - (n+2)(2n+1)\} = 0$ for all θ, but $P(g = 0) = 0$. So the statistic is not complete.

Solution 47 *Two-parameter exponential*

(a) $L = \sigma^{-n}\exp\{-\sum(x_i-\mu)/\sigma\}\mathbf{1}\{\mu \le x_{(1)}\}$ increases in μ up to $x_{(1)}$, so $\hat\mu = X_{(1)}$. Then $\log L = -n\log\sigma - \sum(x_i - x_{(1)})/\sigma$ is maximized at $\hat\sigma = \bar X - X_{(1)}$.

(b) $X_i = \mu+\sigma Z_i$ with $Z_i \overset{\text{iid}}{\sim} \text{Exp}(1)$. The normalized spacings $D_i = (n-i+1)(Z_{(i)}-Z_{(i-1)})$ are i.i.d. Exp(1). So $n(\hat\mu-\mu)/\sigma = D_1 \sim \text{Exp}(1)$, and $n\hat\sigma/\sigma = \sum(Z_i - Z_{(1)}) = \sum_{i\ge 2} D_i \sim \text{Gamma}(n-1,1)$, i.e. $2n\hat\sigma/\sigma \sim \chi^2(2n-2)$, independent of D_1.

(c) $E(n\hat\sigma) = (n-1)\sigma$, so $\tilde\sigma = \frac{n\hat\sigma}{n-1} = \frac{1}{n-1}\sum(X_i - X_{(1)})$ is the UMVUE of σ. Since $EX_{(1)} = \mu+\sigma/n$, the UMVUE of μ is $X_{(1)} - \tilde\sigma/n$.

Solution 48 *Normal with equal mean and variance*

(a) $l(\theta) = -\frac{n}{2}\log\theta - \frac{1}{2\theta}\sum(x_i-\theta)^2 + c$, and

$$\dot l(\theta) = -\frac{n}{2\theta} + \frac{\sum x_i^2}{2\theta^2} - \frac{n}{2} = \frac{n}{2\theta^2}(m_2 - \theta - \theta^2).$$

It is positive for $\theta < \hat\theta$ and negative after, where $\hat\theta$ is the positive root of $\theta^2+\theta-m_2=0$.

(b) $I(\theta) = \frac{1}{\theta} + \frac{1}{2\theta^2} = \frac{2\theta+1}{2\theta^2}$ (mean part plus variance part of a normal), so $\sqrt{n}(\hat\theta-\theta) \xrightarrow{d} N\big(0, \frac{2\theta^2}{2\theta+1}\big)$. (Check by the delta method: $\text{Var}\,X^2 = 2\theta^2+4\theta^3$ and $g'(m_2) = 1/(2\theta+1)$.)

(c) $\bar X$ has variance θ/n and S^2 asymptotic variance $2\theta^2/n$. So $ARE(\bar X,\hat\theta) = \frac{2\theta}{2\theta+1}$ and $ARE(S^2,\hat\theta) = \frac{1}{2\theta+1}$, both below 1.

Solution 49 *Two-parameter uniform*

(a) With $D = \theta_2-\theta_1$, $X_i = \theta_1 + DU_i$ for $U_i \overset{\text{iid}}{\sim} U(0,1)$, so $EX_{(1)} = \theta_1 + \frac{D}{n+1}$ and $EX_{(n)} = \theta_2 - \frac{D}{n+1}$.

(b) $E(X_{(n)} - X_{(1)}) = \frac{n-1}{n+1}D$, so $\frac{n+1}{n-1}(X_{(n)} - X_{(1)})$ is the UMVUE of D. The midpoint has UMVUE $\frac{1}{2}(X_{(1)} + X_{(n)})$. For θ_1: $X_{(1)} - \frac{X_{(n)}-X_{(1)}}{n-1}$, whose mean is $\theta_1 + \frac{D}{n+1} - \frac{D}{n+1} = \theta_1$. All are functions of the complete sufficient statistic.

Solution 50 *Minimal sufficiency* **[Past exam 2024-05 #3]**

$L(\theta;\mathbf{x}) = \mathbf{1}\{x_{(n)} - 1 < \theta < x_{(1)}\}$. For two samples, $L(\theta;\mathbf{x})/L(\theta;\mathbf{y})$ is free of θ iff the two intervals $(x_{(n)}-1, x_{(1)})$ and $(y_{(n)}-1, y_{(1)})$ coincide, i.e. iff $(x_{(1)}, x_{(n)}) = (y_{(1)}, y_{(n)})$. Hence $(X_{(1)}, X_{(n)})$ is minimal sufficient (two-dimensional for a one-dimensional parameter).

Solution 51 *Sufficiency, completeness, UMVUE*

(a) $\prod f(x_i;\theta) = e^{-\sum x_i + n\theta}\mathbf{1}\{x_{(1)} > \theta\} = \big[e^{n\theta}\mathbf{1}\{x_{(1)} > \theta\}\big]e^{-\sum x_i}$; by factorization $X_{(1)}$ is sufficient. $P(X_{(1)} > y) = e^{-n(y-\theta)}$ for $y > \theta$, so $X_{(1)} - \theta \sim \mathrm{Exp}(1/n)$.

(b) If $E_\theta g(X_{(1)}) = \int_\theta^\infty g(y) n e^{-n(y-\theta)} dy = 0$ for all θ, then $\int_\theta^\infty g(y)e^{-ny}dy = 0$ for all θ. Differentiating in θ, $g(\theta)e^{-n\theta} = 0$ for almost every θ, so $g = 0$ a.e. and $P_\theta(g(X_{(1)}) = 0) = 1$.

(c) $EX_{(1)} = \theta + \frac{1}{n}$, so $X_{(1)} - \frac{1}{n}$ is the UMVUE of θ (Lehmann–Scheffé). Also $EX_{(1)}^2 = \operatorname{Var} X_{(1)} + (EX_{(1)})^2 = \frac{1}{n^2} + \theta^2 + \frac{2\theta}{n} + \frac{1}{n^2}$, so

$$E\Big[X_{(1)}^2 - \frac{2}{n}X_{(1)}\Big] = \theta^2 + \frac{2\theta}{n} + \frac{2}{n^2} - \frac{2\theta}{n} - \frac{2}{n^2} = \theta^2,$$

and $X_{(1)}^2 - \frac{2}{n}X_{(1)}$ is the UMVUE of θ^2.

(d) Write $X_i = \theta + Z_i$ with $Z_i \overset{\text{iid}}{\sim} \mathrm{Exp}(1)$; then $S = \sum(Z_i - Z_{(1)})$ does not depend on θ: it is ancillary. $X_{(1)}$ is complete sufficient, so by Basu's theorem they are independent. With $D_i = (n-i+1)(Z_{(i)} - Z_{(i-1)}) \overset{\text{iid}}{\sim} \mathrm{Exp}(1)$ (exponential spacings), $S = \sum_{i=2}^n (Z_{(i)} - Z_{(1)}) = \sum_{i=2}^n D_i \sim \mathrm{Gamma}(n-1, 1)$.

Solution 52 *Basu's theorem*

(a) With σ^2 known, $\bar X$ is complete sufficient for μ. S^2 and $X_{(n)} - X_{(1)}$ do not change when every X_i is shifted, so their laws do not depend on μ: they are ancillary, and Basu's theorem gives independence.

(b) $X_{(n)} - \bar X$ is also location invariant, hence independent of $\bar X$. So $E(X_{(n)} \mid \bar X) = \bar X + E(X_{(n)} - \bar X) = \bar X + \sigma E(Z_{(n)})$ with $Z_i \overset{\text{iid}}{\sim} N(0,1)$: $c_n = EZ_{(n)}$.

(c) $X_1 - \bar X \sim N(0, \sigma^2\frac{n-1}{n})$ is independent of $\bar X$, so $E(X_1^2 \mid \bar X) = \bar X^2 + \sigma^2\frac{n-1}{n}$.

Solution 53 *Rao–Blackwellization*

(a) $X_1 \mid T = t \sim B(t, 1/n)$, so $E(\mathbf{1}\{X_1 = 0\} \mid T) = (1 - 1/n)^T$, the UMVUE of $e^{-\theta}$ (T is complete sufficient).

(b) $E(X_1X_2 \mid T = t) = P(X_1 = X_2 = 1 \mid T = t) = \binom{n-2}{t-2}\big/\binom{n}{t} = \frac{t(t-1)}{n(n-1)}$; the UMVUE of θ^2 is $\frac{T(T-1)}{n(n-1)}$.

(c) Given $X_{(n)} = y$, $X_1 = y$ with probability $\frac{1}{n}$ and otherwise is $U(0, y)$. So $E(2X_1 \mid X_{(n)}) = 2X_{(n)}\big(\frac{1}{n} + \frac{n-1}{2n}\big) = \frac{n+1}{n}X_{(n)}$.

Solution 54 *Cramér–Rao bound*

(a) $I(\theta) = 1/\theta$, so the bound is $\frac{(de^{-\theta}/d\theta)^2}{nI(\theta)} = \frac{\theta e^{-2\theta}}{n}$.

(b) $Es^T = e^{n\theta(s-1)}$, so $E(1 - \frac{1}{n})^{2T} = \exp\{n\theta((1-\frac{1}{n})^2 - 1)\} = e^{-2\theta + \theta/n}$, and

$$\operatorname{Var}\big((1-\tfrac{1}{n})^T\big) = e^{-2\theta}(e^{\theta/n} - 1) > e^{-2\theta}\frac{\theta}{n},$$

since $e^x - 1 > x$ for $x > 0$. No unbiased estimator attains the bound.

(c) By the delta method, $\sqrt{n}(e^{-\bar X} - e^{-\theta}) \xrightarrow{d} N(0, \theta e^{-2\theta})$. Also $n\operatorname{Var}\big((1-\frac{1}{n})^T\big) = e^{-2\theta}n(e^{\theta/n} - 1) \to \theta e^{-2\theta}$: the UMVUE and the MLE are asymptotically equivalent and asymptotically attain the bound.

Solution 55 *UMVUE and the Cramér–Rao bound* **[Past exam 2020-10 #3]**

$\bar{X} \sim N(\mu, \sigma^2/n)$ is complete sufficient, and $Ee^{c\bar{X}} = e^{c\mu + c^2\sigma^2/(2n)}$. So the UMVUE is $e^{c\bar{X} - c^2\sigma^2/(2n)}$, with variance $e^{2c\mu}(e^{c^2\sigma^2/n} - 1)$. The CR bound is $\frac{(ce^{c\mu})^2}{n/\sigma^2} = e^{2c\mu}\frac{c^2\sigma^2}{n}$. Since $e^x - 1 > x$ for $x > 0$, the bound is not attained.

Solution 56 *UMVUE of a normal probability* **[Past exam 2023-04 #4]**

$Eu(X_1) = P(X_1 \le c) = \Phi(c - \theta)$. $\bar{X}$ is complete sufficient, and $X_1 - \bar{X} \sim N(0, 1 - \frac{1}{n})$ is independent of $\bar{X}$, so

$$E\{u(X_1) \mid \bar{X}\} = P(X_1 - \bar{X} \le c - \bar{X} \mid \bar{X}) = \Phi\Big(\frac{c - \bar{X}}{\sqrt{1 - 1/n}}\Big),$$

the UMVUE by Lehmann–Scheffé.

Solution 57 *Estimating a Poisson zero probability* **[Past exam 2023-10 #1]**

(1) By the delta method, $\sqrt{n}(\hat{\theta}^{(1)} - \theta) \xrightarrow{d} N(0, \lambda e^{-2\lambda})$. (2) $\hat{\theta}^{(2)}$ is a mean of i.i.d. Bernoulli$(e^{-\lambda})$, so $\sqrt{n}(\hat{\theta}^{(2)} - \theta) \xrightarrow{d} N(0, e^{-\lambda}(1 - e^{-\lambda}))$. (3) $ARE(\hat{\theta}^{(2)}, \hat{\theta}^{(1)}) = \frac{\lambda e^{-2\lambda}}{e^{-\lambda}(1 - e^{-\lambda})} = \frac{\lambda}{e^{\lambda} - 1} < 1$, so $\hat{\theta}^{(1)}$ (the MLE) is more efficient.

Solution 58 *UMVUE for a uniform region*

(a) The square $\{|x| + |y| < w\}$ has area $2w^2$, so $P(W_1 \le w) = (w/\theta)^2$ on $(0, \theta)$. The joint pdf of the sample is $(2\theta^2)^{-n}\mathbf{1}\{M < \theta\}$, so M is sufficient by factorization.

(b) M has cdf $(m/\theta)^{2n}$ and pdf $2nm^{2n-1}/\theta^{2n}$. If $\int_0^\theta g(m)m^{2n-1}dm = 0$ for all $\theta > 0$, differentiating gives $g(\theta)\theta^{2n-1} = 0$, so $g = 0$ a.e.

(c) $EM = \frac{2n}{2n+1}\theta$ and $EM^2 = \frac{2n}{2n+2}\theta^2$. By Lehmann–Scheffé the UMVUEs are $\frac{2n+1}{2n}M$ for θ and $\frac{2(n+1)}{n}M^2$ for $2\theta^2$.

Solution 59 *UMVUE for a triangle* **[Past exam 2021-04 #5]**

$W = X + Y$ has $P(W \le w) = (w/\theta)^2$. The joint pdf is $(2/\theta^2)^n\mathbf{1}\{M < \theta\}$ with $M = \max W_i$, which is sufficient and (as in Problem 58) complete, with cdf $(m/\theta)^{2n}$ and $EM = \frac{2n}{2n+1}\theta$. The UMVUE is $T = \frac{2n+1}{2n}M$, with pdf

$$f_T(t) = \frac{2n\,t^{2n-1}}{(c\theta)^{2n}}, \qquad 0 < t < c\theta, \quad c = \frac{2n+1}{2n}.$$

Solution 60 *UMVUE for a disk* **[Past exam 2021-10 #4]**

$R_i^2 = X_i^2 + Y_i^2$ has $P(R^2 \le s) = \pi s/(\pi r^2) = s/\theta$, so $R_i^2 \overset{\text{iid}}{\sim} U(0, \theta)$, and $M = \max R_i^2$ is complete sufficient. The UMVUE is $T = \frac{n+1}{n}M$, with pdf $f_T(t) = nt^{n-1}/(c\theta)^n$ on $(0, c\theta)$, $c = \frac{n+1}{n}$.

Solution 61 *UMVUE of a ratio of uniform scales* **[Past exam 2022-04 #1]**

$(X_{(m)}, Y_{(n)})$ is complete sufficient (product of independent complete families). $EX_{(m)} = \frac{m}{m+1}\theta$, and for $n \geq 2$, $EY_{(n)}^{-1} = \int_0^\eta y^{-1} \frac{ny^{n-1}}{\eta^n} dy = \frac{n}{(n-1)\eta}$. By independence the UMVUE is

$$\frac{m+1}{m} \cdot \frac{n-1}{n} \cdot \frac{X_{(m)}}{Y_{(n)}}.$$

Solution 62 *Kullback–Leibler divergence*

(a) Normal: $E_{\theta_0} \frac{1}{2}\{(X-\theta_1)^2 - (X-\theta_0)^2\} = \frac{1}{2}(\theta_1 - \theta_0)^2$. Poisson: $\log \frac{f_0}{f_1} = x \log \frac{\theta_0}{\theta_1} - \theta_0 + \theta_1$, so $KL = \theta_0 \log \frac{\theta_0}{\theta_1} - \theta_0 + \theta_1$.

(b) $-KL = E_0 \log \frac{f_1}{f_0} \leq E_0\big(\frac{f_1}{f_0} - 1\big) = \int_{f_0>0} f_1 - 1 \leq 0$, with equality iff $f_1/f_0 = 1$ with probability one.

(c) By the LLN, $\frac{1}{n} l_n(\theta) \to E_{\theta_0} \log f(X;\theta)$ for each θ, and by (b) this limit is uniquely maximized at θ_0. The maximizer of $\frac{1}{n} l_n$, the MLE, should then approach θ_0 (rigorous under uniform convergence or concavity).

Solution 63 *Censored data*

(a) An uncensored observation contributes $\theta^{-1} e^{-z/\theta}$ and a censored one $P(X > c) = e^{-c/\theta}$. Hence $L(\theta) = \theta^{-D} e^{-T/\theta}$, and (D, T) is sufficient.

(b) $l = -D \log \theta - T/\theta$, so $\dot{l} = -D/\theta + T/\theta^2$, giving $\hat{\theta} = T/D$ (total time on test divided by the number of failures).

(c) $ED = nP_\theta$ and $E \min(X, c) = \int_0^c P(X > x) dx = \theta P_\theta$. With $\ddot{l} = D/\theta^2 - 2T/\theta^3$, $I_n(\theta) = -E\ddot{l} = -\frac{nP_\theta}{\theta^2} + \frac{2n\theta P_\theta}{\theta^3} = \frac{nP_\theta}{\theta^2}$. So $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \theta^2/P_\theta)$. Without censoring the variance would be θ^2; censoring inflates it by the factor $1/P_\theta$, the inverse of the probability of observing a failure.

Solution 64 *MLE on a restricted parameter space* **[Past exam 2021-04 #3]**

$l(\theta) = -n \log \theta - n\bar{x}/\theta$ increases on $(0, \bar{x})$ and decreases afterwards, so on $[1, \infty)$ the MLE is $\hat{\theta} = \max(\bar{X}, 1)$. If $\theta > 1$, $P(\bar{X} < 1) \to 0$, so $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \theta^2)$. If $\theta = 1$, $\sqrt{n}(\hat{\theta} - 1) = \{\sqrt{n}(\bar{X} - 1)\}^+ \xrightarrow{d} Z^+$, $Z \sim N(0,1)$: half the mass sits at 0.

E. Hypothesis testing

Solution 65 *Most powerful and UMP tests*

(a) $P(\log X > t) = P(X > e^t) = e^{-\theta t}$, so $\log X_i \overset{\text{iid}}{\sim} \text{Exp}(1/\theta)$, $T \sim \text{Gamma}(n, 1/\theta)$ and $2\theta T \sim \text{Gamma}(n, 2) = \chi^2(2n)$.

(b) $\dfrac{\prod f(x_i;\theta_1)}{\prod f(x_i;1)} = \theta_1^n \exp\{-(\theta_1 - 1)T\}$, decreasing in T for $\theta_1 > 1$. By Neyman–Pearson, the MP test rejects for $T \le c$. Under $\theta = 1$, $2T \sim \chi^2(2n)$, so $c = \frac{1}{2}\chi^2_{1-\alpha}(2n)$: reject when $2T \le \chi^2_{1-\alpha}(2n)$.

(c) The test does not depend on θ_1, so it is UMP for $\theta = 1$ vs $\theta > 1$. Its power is

$$\gamma(\theta) = P_\theta\big(2T \le \chi^2_{1-\alpha}(2n)\big) = P\big(\chi^2(2n) \le \theta\chi^2_{1-\alpha}(2n)\big),$$

increasing in θ. So $\max_{\theta \le 1} \gamma(\theta) = \gamma(1) = \alpha$, and any level-$\alpha$ test of $\theta \le 1$ is level α for $\theta = 1$; hence the test is UMP for $H_0 : \theta \le 1$. (This is also the one-parameter exponential family result with $g(\theta) = -\theta$ decreasing in θ, applied to T.)

(d) Reject when $2T \le 10.851$, i.e. $\sum \log X_i \le 5.43$. Power at $\theta = 2$: $P(\chi^2(20) \le 21.70) \approx 0.64$.

Solution 66 *Power and sample size*

(a) Reject when $\bar{X} \ge 10 + 1.645 \cdot 4/\sqrt{n}$; the power is $\gamma(\mu) = \Phi\big(\sqrt{n}(\mu - 10)/4 - 1.645\big)$.

(b) $\sqrt{n} \cdot \frac{2}{4} - 1.645 \ge 1.282$ gives $n \ge \{2(1.645 + 1.282)\}^2 = 34.3$: $n = 35$.

(c) $\gamma(11) = \Phi(\sqrt{35}/4 - 1.645) = \Phi(-0.166) = 0.434$.

(d) The MP test against $\mu_1 > 10$ rejects for large $\bar{X}$, and against $\mu_1 < 10$ for small $\bar{X}$ (Neyman–Pearson, with uniqueness of MP tests). No single test is MP against both.

Solution 67 *Tests for a normal variance*

(a) The density is $\exp\{-T/(2\sigma^2) - \frac{n}{2}\log(2\pi\sigma^2)\}$ in $T = \sum(x_i - \mu)^2$, with $-1/(2\sigma^2)$ increasing in σ^2. The UMP test rejects for large T: $T \ge \sigma_0^2\chi^2_\alpha(n)$, since $T/\sigma_0^2 \sim \chi^2(n)$ at σ_0^2. Power: $\gamma(\sigma^2) = P\big(\chi^2(n) \ge \sigma_0^2\chi^2_\alpha(n)/\sigma^2\big)$.

(b) With $\hat{\sigma}^2 = \frac{1}{n}\sum(x_i - \bar{x})^2$, $l(\sigma^2) = -\frac{n}{2}\log\sigma^2 - \frac{n\hat{\sigma}^2}{2\sigma^2}$ at $\mu = \bar{x}$, so

$$2\{l(\hat{\sigma}^2) - l(\sigma_0^2)\} = n\Big(\frac{\hat{\sigma}^2}{\sigma_0^2} - 1 - \log\frac{\hat{\sigma}^2}{\sigma_0^2}\Big) = n(v - 1 - \log v).$$

This is convex in v with minimum at $v = 1$, so we reject for $v \le c_1$ or $v \ge c_2$, where $nv = (n-1)S^2/\sigma_0^2 \sim \chi^2(n-1)$ under H_0.

Solution 68 *Likelihood ratio test*

(a) Unrestricted MLEs: $\hat{\theta}_1 = \bar{X}$, $\hat{\theta}_2 = \bar{Y}$; under H_0: $\hat{\theta}_0 = (n\bar{X} + m\bar{Y})/(n+m)$. Since $l(\theta_1, \theta_2) = -n\log\theta_1 - n\bar{x}/\theta_1 - m\log\theta_2 - m\bar{y}/\theta_2$, the terms $n\bar{x}/\hat{\theta}$ etc. sum to $n + m$ in both cases, and

$$2\log\Lambda = 2\big\{(n+m)\log\hat{\theta}_0 - n\log\bar{X} - m\log\bar{Y}\big\}.$$

With $B = n\bar{X}/(n\bar{X} + m\bar{Y})$, $\bar{X} = \frac{(n+m)\hat{\theta}_0}{n}B$ and $\bar{Y} = \frac{(n+m)\hat{\theta}_0}{m}(1 - B)$, so

$$2\log\Lambda = -2\{n\log B + m\log(1-B)\} + 2\{n\log\tfrac{n}{n+m} + m\log\tfrac{m}{n+m}\}.$$

The function $-n\log b - m\log(1-b)$ is convex with minimum at $b = n/(n+m)$, so the LRT rejects when $B \le c_1$ or $B \ge c_2$, where $c_1 < \frac{n}{n+m} < c_2$ give equal values of the statistic and total size α.

(b) Under H_0 (common θ), $n\bar{X}/\theta \sim \text{Gamma}(n,1)$ and $m\bar{Y}/\theta \sim \text{Gamma}(m,1)$ are independent, so $B \sim \text{Beta}(n,m)$, free of θ. Equivalently $\frac{2n\bar{X}/\theta}{2n} \Big/ \frac{2m\bar{Y}/\theta}{2m} = \bar{X}/\bar{Y} \sim F(2n, 2m)$.

(c) $\dim\Omega - \dim\Omega_0 = 2 - 1 = 1$, so $2\log\Lambda \xrightarrow{d} \chi^2(1)$ under H_0.

Solution 69 *LRT for a uniform scale*

(a) $L(\theta) = \theta^{-n}\mathbf{1}\{\theta \ge x_{(n)}\}$ and $\hat{\theta} = X_{(n)}$, so $\Lambda = L(\hat{\theta})/L(\theta_0) = (\theta_0/X_{(n)})^n$ if $X_{(n)} \le \theta_0$ and ∞ otherwise. We reject when $X_{(n)} > \theta_0$ or $X_{(n)} \le c$; the size is $P_{\theta_0}(X_{(n)} \le c) = (c/\theta_0)^n = \alpha$, so $c = \theta_0\alpha^{1/n}$.

(b) Under H_0, $U = (X_{(n)}/\theta_0)^n \sim U(0,1)$, and $2\log\Lambda = -2\log U \sim \text{Exp}(2) = \chi^2(2)$. The usual $\chi^2(1)$ limit requires the regularity conditions; here the support depends on θ, and the MLE converges at rate n rather than $\sqrt{n}$.

Solution 70 *Non-uniqueness of UMP tests*

For $\theta_1 > \theta_0$, $p_{\theta_1}/p_{\theta_0} = (\theta_0/\theta_1)^n =: k$ on $\{x_{(n)} \le \theta_0\}$ and ∞ on $\{\theta_0 < x_{(n)} \le \theta_1\}$. Any test that rejects on $\{x_{(n)} > \theta_0\}$ and has size α at θ_0 is of Neyman–Pearson form with this k, hence MP. Both tests reject when $x_{(n)} > \theta_0$. At θ_0, $E\phi_1 = 1 - (1-\alpha) = \alpha$ and $E\phi_2 = 0 + \alpha = \alpha$. Their powers for $\theta > \theta_0$ are

$$1 - (1-\alpha)\Big(\frac{\theta_0}{\theta}\Big)^n \quad \text{and} \quad 1 - \Big(\frac{\theta_0}{\theta}\Big)^n + \alpha\Big(\frac{\theta_0}{\theta}\Big)^n,$$

which are equal. For $\theta \le \theta_0$ both powers are at most α: for ϕ_2 it is exactly α, and for ϕ_1 it is $\max\{0, 1-(1-\alpha)(\theta_0/\theta)^n\} \le \alpha$. So both are UMP for the composite null.

Solution 71 *UMP test for a uniform location*

(a) $p_0 = \mathbf{1}\{0 < x_{(1)}, x_{(n)} < 1\}$ and $p_{\theta_1} = \mathbf{1}\{\theta_1 < x_{(1)}, x_{(n)} < \theta_1 + 1\}$. The ratio is ∞ where $x_{(n)} \ge 1$ (only $p_{\theta_1} > 0$), equals 1 where $\theta_1 < x_{(1)}$ and $x_{(n)} < 1$, and equals 0 where $x_{(1)} \le \theta_1$.

(b) Size: $P_0(X_{(1)} \ge c) = (1-c)^n = \alpha$, and $P_0(X_{(n)} \ge 1) = 0$. Fix θ_1. If $\theta_1 < c$, the test rejects everywhere the ratio is ∞, never where it is 0, and on part of the region where it equals $k = 1$. This is the Neyman–Pearson form with $k = 1$ and size α, so the test is MP. If $\theta_1 \ge c$, its power at θ_1 is 1, which cannot be beaten. The test does not depend on θ_1, so it is UMP.

(c) For $0 < \theta < c$, $1 - P_\theta(X_{(n)} < 1, X_{(1)} < c) = 1 - \{(1-\theta)^n - (1-c)^n\} = 1 - (1-\theta)^n + \alpha$. For $\theta \ge c$ the power is 1.

Solution 72 *Non-uniqueness of MP tests* **[Past exam 2024-10 #1]**

With $k = (\theta_0/\theta_1)^n$, $p_{\theta_1} = kp_{\theta_0}$ on $\{x_{(n)} \le \theta_0\}$, and $p_{\theta_0} = 0 < p_{\theta_1}$ on $\{\theta_0 < x_{(n)} \le \theta_1\}$. For any ψ with $E_{\theta_0}\psi \le \alpha$,

$$E_{\theta_1}\phi - E_{\theta_1}\psi = \int_{x_{(n)} \le \theta_0} (\phi - \psi)kp_{\theta_0} + \int_{x_{(n)} > \theta_0} (1-\psi)p_{\theta_1} \ge k(\alpha - E_{\theta_0}\psi) \ge 0.$$

Solution 73 *UMP test for a uniform scale* **[Past exam 2021-04 #4]**

Reject when $X_{(n)} > \theta_0$ or $X_{(n)} \le \theta_0\alpha^{1/n}$; the size is $0 + \alpha$. For $\theta_1 > \theta_0$ the test is MP by Problem 70 (it rejects on $\{x_{(n)} > \theta_0\}$ and has size α). For $\theta_1 < \theta_0$ the ratio $p_{\theta_1}/p_{\theta_0}$ is $(\theta_0/\theta_1)^n = k$ on $\{x_{(n)} \le \theta_1\}$ and 0 on $(\theta_1, \theta_0]$. If $\theta_0\alpha^{1/n} \le \theta_1$, the rejection region lies where the ratio equals k, so the test is of NP form and MP. Otherwise its power at θ_1 is $P_{\theta_1}(X_{(n)} \le \theta_0\alpha^{1/n}) = 1$. Hence it is UMP.

Solution 74 *UMP test for a Pareto scale* **[Past exam 2022-10 #4]**

$X_{(1)}$ is sufficient with $P(X_{(1)} > t) = (a/t)^{n\theta}$, $t > a$. Reject when $X_{(1)} < a_0$ or $X_{(1)} \ge a_0\alpha^{-1/(n\theta)}$; the size is $0 + (a_0/(a_0\alpha^{-1/(n\theta)}))^{n\theta} = \alpha$. The argument is that of Problem 35 applied to $1/X_i$, a scale family on $(0, 1/a)$. For $a_1 > a_0$ the ratio is 0 on $(a_0, a_1]$ and constant on $x_{(1)} > a_1$, and the test is MP or has power 1. For $a_1 < a_0$ the ratio is ∞ on $(a_1, a_0]$, where the test rejects, and constant elsewhere. So the test is UMP.

Solution 75 *UMP test for a logistic location* **[Past exam 2022-04 #2]**

For $\theta_0 < \theta_1$,

$$\frac{f(x;\theta_1)}{f(x;\theta_0)} = e^{-(\theta_1-\theta_0)}\Big(\frac{1+e^{x-\theta_0}}{1+e^{x-\theta_1}}\Big)^2.$$

The derivative of the bracket has the sign of $e^{x-\theta_0} - e^{x-\theta_1} > 0$, so the family has monotone likelihood ratio in x. By Neyman–Pearson the MP test of θ_0 vs θ_1 rejects for $X > C$. Its power is increasing in θ (location family), so the test is UMP for $\theta \le 0$ vs $\theta > 0$ (as in Theorem 9.3 of the notes). Size: $P_0(X > C) = 1/(1 + e^C) = \alpha$, so $C = \log\frac{1-\alpha}{\alpha}$.

Solution 76 *UMP test for a Weibull model* **[Past exam 2023-04 #5]**

(1) $P(X^c > t) = P(X > t^{1/c}) = e^{-\lambda t}$: X^c is exponential with rate λ. (2) The joint pdf is $\lambda^n \exp(-\lambda T)\prod c x_i^{c-1}$ with $T = \sum X_i^c$, an exponential family in which small λ makes large T likely. The UMP test of $\lambda \ge \lambda_0$ vs $\lambda < \lambda_0$ rejects for large T. Since $2\lambda_0 T \sim \chi^2(2n)$ at λ_0, it rejects when $2\lambda_0 \sum X_i^c \ge \chi^2_\alpha(2n)$.

Solution 77 *Reversing a UMP test* **[Past exam 2023-10 #2]**

The UMP test has NP form: $\phi = 1$ where $f_1 > kf_0$ and $\phi = 0$ where $f_1 < kf_0$, for some $k > 0$. Then $1 - \phi = 1$ where $f_0 > k^{-1}f_1$ and 0 where $f_0 < k^{-1}f_1$: NP form for f_1 vs f_0. Its size under f_1 is $1 - \beta$ and its power under f_0 is $1 - \alpha$. So $\phi^* = 1 - \phi$ is the UMP size-$(1 - \beta)$ test.

Solution 78 *Wald, score and LRT*

(a) $\hat{p} = 0.6$.

- LRT: $2\{60 \log 1.2 + 40 \log 0.8\} = 2(10.939 - 8.926) = 4.03$.
- Wald: $\frac{(0.6-0.5)^2}{0.6 \cdot 0.4/100} = 4.17$.
- Score: $\frac{(60-50)^2}{100 \cdot 0.25} = 4.00$.

(b) Each is asymptotically $\chi^2(1)$ under H_0. All exceed 3.84, so H_0 is rejected (the score statistic only barely).

(c) The score statistic uses only p_0. The score confidence set is $\{p : (60-100p)^2 \le 3.84 \cdot 100p(1-p)\}$, an interval found by solving the quadratic.

Solution 79 *LRT on a restricted parameter space*

(a) $l(\mu) = -\frac{n}{2}(\bar{x} - \mu)^2 + \text{const}$ is maximized over $\mu \ge 0$ at $\hat{\mu} = \bar{X}^+$. So $2 \log \Lambda = 2\{l(\hat{\mu}) - l(0)\} = n\{\bar{X}^2 - (\bar{X} - \bar{X}^+)^2\} = n(\bar{X}^+)^2$.

(b) Under H_0, $Z = \sqrt{n}\bar{X} \sim N(0,1)$ and $2 \log \Lambda = (Z^+)^2$, which is 0 with probability $\frac{1}{2}$ and otherwise distributed as Z^2 given $Z > 0$, i.e. $\chi^2(1)$. Hence $P(2 \log \Lambda > c) = \frac{1}{2}P(\chi^2(1) > c) = \alpha$ gives $c = \chi^2_{2\alpha}(1) = z_\alpha^2$: the test rejects when $\sqrt{n}\bar{X} > z_\alpha$. The naive critical value $\chi^2_\alpha(1)$ gives actual size $\alpha/2$.

(c) For $\mu > 0$, $P(\bar{X} < 0) \to 0$, so $\sqrt{n}(\hat{\mu} - \mu) \xrightarrow{d} N(0,1)$. For $\mu = 0$, $\sqrt{n}\hat{\mu} = (\sqrt{n}\bar{X})^+ \sim Z^+$ exactly: a point mass $\frac{1}{2}$ at 0 plus the positive half of $N(0,1)$. The MLE is not asymptotically normal at the boundary.

Solution 80 *LRT on a restricted parameter space* **[Past exam 2020-10 #2]**

The MLE over Ω is $\hat{\mu}_i = X_i^+$, so

$$2 \log \Lambda = \sum_i \{X_i^2 - (X_i - X_i^+)^2\} = (X_1^+)^2 + (X_2^+)^2.$$

Reject when this is at least c. Under H_0, each $(X_i^+)^2$ is 0 or $\chi^2(1)$ with probability $\frac{1}{2}$ each, and the number of positive X_i is $B(2, \frac{1}{2})$. Hence the null law is $\frac{1}{4}\delta_0 + \frac{1}{2}\chi^2(1) + \frac{1}{4}\chi^2(2)$, and c solves $\frac{1}{2}P(\chi^2(1) \ge c) + \frac{1}{4}P(\chi^2(2) \ge c) = \alpha$.

Solution 81 *LRT for a two-parameter exponential* **[Past exam 2021-10 #3]**

Unrestricted: $\hat{\mu} = X_{(1)}$ and $\hat{\sigma} = \bar{X} - X_{(1)}$. Under H_0 (possible only if $X_{(1)} \ge \mu_0$): $\hat{\sigma}_0 = \bar{X} - \mu_0 = \hat{\sigma} + (X_{(1)} - \mu_0)$. In both cases the exponential factor equals e^{-n}, so $\Lambda = (\hat{\sigma}_0/\hat{\sigma})^n = \{1 + (X_{(1)} - \mu_0)/\hat{\sigma}\}^n$. Reject when $X_{(1)} < \mu_0$ or $(X_{(1)} - \mu_0)/\hat{\sigma}$ is large. Under H_0, $2n(X_{(1)} - \mu_0)/\sigma \sim \chi^2(2)$ independently of $2n\hat{\sigma}/\sigma \sim \chi^2(2n-2)$ (Problem 47). So the test rejects when $X_{(1)} < \mu_0$ or

$$\frac{(n-1)(X_{(1)} - \mu_0)}{\hat{\sigma}} \ge F_\alpha(2, 2n-2).$$

F. Interval estimation and Bayesian inference

Solution 82 *Pivots and confidence intervals*

(a) $2n\bar{X}/\theta \sim \chi^2(2n)$ gives $\Big[\dfrac{2n\bar{X}}{\chi^2_{\alpha/2}(2n)}, \dfrac{2n\bar{X}}{\chi^2_{1-\alpha/2}(2n)}\Big]$.

(b) $W = X_{(n)}/\theta$ has cdf w^n, so the coverage is $P(a \le W \le b) = b^n - a^n = 1 - \alpha$, and the length is $X_{(n)}(1/a - 1/b)$. Treating $a = a(b)$ through $a^n = b^n - (1-\alpha)$, $a'(b) = b^{n-1}/a^{n-1}$, and

$$\frac{d}{db}\Big(\frac{1}{a} - \frac{1}{b}\Big) = -\frac{b^{n-1}}{a^{n+1}} + \frac{1}{b^2} < 0,$$

because $a < b$. The length decreases in b, so $b = 1$ and $a = \alpha^{1/n}$ are optimal.

(c) $\Big[\sqrt{(n-1)S^2/\chi^2_{\alpha/2}(n-1)},\ \sqrt{(n-1)S^2/\chi^2_{1-\alpha/2}(n-1)}\Big]$.

Solution 83 *Bayes and minimax*

(a) $\pi(\theta \mid \mathbf{x}) \propto e^{-n\theta}\theta^T \cdot e^{-\theta/\beta} = \theta^T e^{-\theta/s}$, so $\theta \mid \mathbf{x} \sim \text{Gamma}(T+1, s)$.

(b) The posterior mean is $s(T+1) = \dfrac{n}{n+1/\beta}\bar{X} + \dfrac{1/\beta}{n+1/\beta}\beta$, a weighted average of $\bar{X}$ and the prior mean β.

(c) The posterior expected loss is $\rho(a) = a^2 E(\theta^{-1} \mid \mathbf{x}) - 2a + E(\theta \mid \mathbf{x})$. For $T \ge 1$, $E(\theta^{-1} \mid \mathbf{x}) = \frac{1}{sT}$, so ρ is minimized at $a^* = sT$ with $\rho(a^*) = s(T+1) - sT = s$. For $T = 0$, $E(\theta^{-1} \mid \mathbf{x}) = \infty$, so only $a = 0 = sT$ has finite loss, $\rho(0) = E(\theta \mid \mathbf{x}) = s$. In all cases the Bayes estimator is sT and its posterior expected loss is s, so its Bayes risk is $r(\pi_\beta) = s = (n + 1/\beta)^{-1}$.

(d) $\bar{X}$ has risk $E\big[(\bar{X} - \theta)^2/\theta\big] = \frac{\theta/n}{\theta} = \frac{1}{n}$ for all θ. For any estimator δ, $\max_\theta R_\delta(\theta) \ge r(\pi_\beta, \delta) \ge (n + 1/\beta)^{-1}$ for every β; letting $\beta \to \infty$, $\max_\theta R_\delta \ge \frac{1}{n} = \max_\theta R_{\bar{X}}$. So $\bar{X}$ is minimax (limit of Bayes estimators, least favourable sequence of priors).

Solution 84 *Bayesian prediction and risk*

(a) $P(X_{n+1} = 1 \mid \mathbf{x}) = E(\theta \mid \mathbf{x}) = \frac{t+\alpha}{n+\alpha+\beta}$; with a uniform prior and $t = n$ this is $\frac{n+1}{n+2}$ (Laplace's rule of succession).

(b) The posterior variance $v = (n/\sigma^2 + 1/\tau^2)^{-1}$ does not depend on $\mathbf{x}$, so the Bayes risk of the posterior mean is $E\{\text{Var}(\theta \mid \mathbf{X})\} = v < \sigma^2/n$. The Bayes risk of $\bar{X}$ is its constant MSE σ^2/n.

(c) The posterior mean is $w\bar{X} + (1-w)\mu_0$ with $w = \frac{n/\sigma^2}{n/\sigma^2 + 1/\tau^2} \in (0,1)$. Its MSE is $w^2\frac{\sigma^2}{n} + (1-w)^2(\theta - \mu_0)^2$, which exceeds $\frac{\sigma^2}{n}$ iff $(\theta - \mu_0)^2 > \frac{(1+w)\sigma^2}{(1-w)n}$.

G. Nonparametric methods

Solution 85 *Signed rank test*

(a) Let D_j be the index of the observation whose $|X|$ has rank j. Then $W^+ = \sum_{j=1}^n j\,\mathbf{1}\{X_{D_j} > 0\}$. Since the sign vector is independent of the ranks and its entries are i.i.d. fair, the signs attached to ranks $1, \dots, n$ are again i.i.d. fair, so $W^+ \overset{d}{=} \sum jB_j$.

(b) $EW^+ = \frac{1}{2}\sum j = \frac{n(n+1)}{4}$ and $\operatorname{Var} W^+ = \frac{1}{4}\sum j^2 = \frac{n(n+1)(2n+1)}{24}$. For $n = 4$, W^+ is the sum of a uniformly random subset of $\{1, 2, 3, 4\}$ (16 subsets):

w	0	1	2	3	4	5	6	7	8	9	10
$16\,P(W^+ = w)$	1	1	1	2	2	2	2	2	1	1	1

(c) Order by absolute value: $|Y_{(1)}| < \cdots < |Y_{(n)}|$. Then $W^+ = \sum_i i\mathbf{1}\{Y_{(i)} > 0\} = \sum_i \mathbf{1}\{Y_{(i)} > 0\} + \sum_{j<i} \mathbf{1}\{Y_{(i)} > 0\}$, and for $j < i$, $Y_{(i)} > 0 \iff Y_{(i)} + Y_{(j)} > 0$ since $|Y_{(j)}| < |Y_{(i)}|$. Hence $W^+ = \sum_{i \le j} \mathbf{1}\{X_i + X_j > 0\}$.

(d) Uniform on $(-1, 1)$: $\int f^2 = \frac{1}{2}$, $\sigma^2 = \frac{1}{3}$, so $ARE = 12 \cdot \frac{1}{3} \cdot \frac{1}{4} = 1$. Double exponential: $\int f^2 = \int \frac{1}{4}e^{-2|x|} = \frac{1}{4}$, $\sigma^2 = 2$, so $ARE = 12 \cdot 2 \cdot \frac{1}{16} = \frac{3}{2}$.

Solution 86 *Signed rank statistic* **[Past exam 2025-04 #3]**

(1), (3): see Problem 85(a), using $W_n = 2W_n^+ - n(n+1)/2$. (2) With $\sigma_n^2 = \sum j^2$, the cgf of W_n/σ_n is $\sum_j \log\cosh(jt/\sigma_n) = \frac{t^2}{2} + O(\sum_j j^4/\sigma_n^4)$, and $\sum j^4/\sigma_n^4 = O(n^5/n^6) \to 0$. So the cgf tends to $t^2/2$.

Solution 87 *Wilcoxon rank sum*

(a) Under H_0 the rank vector is uniform over permutations, so each rank is uniform on $\{1, \dots, N\}$: mean $\frac{N+1}{2}$, variance $\frac{N^2-1}{12}$. All N ranks sum to a constant, so $0 = N \operatorname{Var} R + N(N-1)c$, giving $c = -\frac{N+1}{12}$.

(b) $EW = \frac{n(N+1)}{2}$ and

$$\operatorname{Var} W = n\frac{N^2-1}{12} - n(n-1)\frac{N+1}{12} = \frac{n(N+1)(N-n)}{12} = \frac{mn(N+1)}{12}.$$

(c) The $\binom{6}{3} = 20$ subsets of $\{1, \dots, 6\}$ are equally likely; sums ≥ 14 come only from $\{4, 5, 6\}$ and $\{3, 5, 6\}$. So $P(W \ge 14) = 2/20 = 0.1$.